



Dow HFDD-4201 VC EXP1

Cross-linkable Polyethylene Compound for Power Cable Insulation

Overview

DOW HFDD-4201 VC EXP1 is a cross-linkable, low-density polyethylene insulation compound formulated for the insulation of power cables. HFDD-4201 VC EXP1 provides high thermal stability with long term storage stability and enhanced scorch resistance. The HFDD-4201 VC has been designed to achieve exceptional crosslinking performance while generating an industry leading low crosslink byproduct level that enables significantly faster degassing compared to other XLPEs.

DOW HFDD-4201 VC EXP1 is recommended for the insulation of power transmission and distribution cables rated at less than 150kV

Specifications

Cables insulated with HFDD-4201 VC EXP1, using sound commercial manufacturing practice, would be expected to meet the latest editions of the following specifications and regulations:

- IEC 60502, 60840
- ANSI/ICEA*: S-94-649, S-97-682, S-93-639/NEMA WC74
- GB/T 12706
- BSI BS 6622, 7870-4
- CENELEC HD 620 S2

*Applicable to USA manufactured product only

The properties reported are typical properties only and are not meant to be construed as specifications. Users should confirm results by their own tests.

Physical	Nominal Value	(SI)	Nominal Value	(English)	Test Method
Density (73°F (23°C))	0.921	g/cm ³	0.921	g/cm ³	ISO 1183
Moisture (73°F (23°C))	< 200	ppm	< 200	ppm	Internal
Mechanical	Nominal Value	(SI)	Nominal Value	(English)	Test Method
Tensile Strength	18.0	MPa	2600	psi	IEC 60811-1-1
Tensile Elongation (Break)	480	%	480	%	IEC 60811-1-1
Thermal	Nominal Value	(SI)	Nominal Value	(English)	Test Method
Hot set (0.2 MPa, 200 °C 392 °F)					IEC 811-8-1
Elongation under Load	< 100	%	< 100	%	
Permanent Deformation	< 5	%	< 5	%	
Aging	Nominal Value	(SI)	Nominal Value	(English)	Test Method
Change in Tensile Properties (after 7 days at 150 °C 320 °F)	< 25	%	< 25	%	IEC 60811-1-2
Electrical	Nominal Value	(SI)	Nominal Value	(English)	Test Method
Volume resistivity (73°F (23°C))	> 1.0 × 10 ⁺¹⁶	ohm·cm	> 1.0 × 10 ⁺¹⁶	ohm·cm	IEC 62631
Electric Strength	> 30	kV/mm	> 760	V/mil	IEC 60243-1
Dielectric Constant (73°F (23°C))	< 2.30		< 2.30		IEC 62631
Dissipation Factor (73°F (23°C))	< 3.0 × 10 ⁻⁴		< 3.0 × 10 ⁻⁴		IEC 62631

Additional information

Nominal property values above represent tests on molded stress-relieved slabs. Cure times were 15 minutes at 175°C.

Cleanliness Requirements

DOW HFDD-4201 VC EXP1 meets high standards for cleanliness (Very Clean) established for an unfilled, cross-linkable cable insulation compound. Throughout the production process, the product is tested to ensure a high level of cleanliness. Extruded tapes are scanned by an automatic inspection system in a clean room. The purity data is managed using an acceptance sampling plan which ensures that the product meets Dow very-clean standards.

Storage

The environment or conditions of storage greatly influences the recommended storage time. Storage under extreme conditions may affect the quality, processing, or performance of the product. Storage should be in accordance with good manufacturing practices. The recommended storage conditions are dry conditions with temperatures between 10°C and 30°C (50°F and 86°F). When stored under these conditions, the product may be used by the customer for up to one year from the date of sale or two years from the date of manufacture, whichever comes first. It is recommended that the practice of using the product on a first-in / first-out basis be established.

Packaging

Dow HFDD-4201 EHVP1 can be delivered in different packaging types dependent on specific material handling needs. This includes 500kg UNICLEAN™ octabins or in 1000kg bottom unloading octabins.

Extrusion Notes

Extrusion Parameter	Nominal Value	(SI)	Nominal Value	(English)
Melt Temperature	115 to 140	°C	239 to 284	°F

DOW HFDD-4201 VC EXP1 provides outstanding output rates over a broad range of conditions. For optimum results, melt extrusion temperatures of 115 °C to ≤ 135 °C are recommended, although higher melt temperatures are possible on certain equipment with due care. Specific recommendations for processing conditions can be determined when the application and type of processing equipment are known.

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Published: 2019-10-01

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